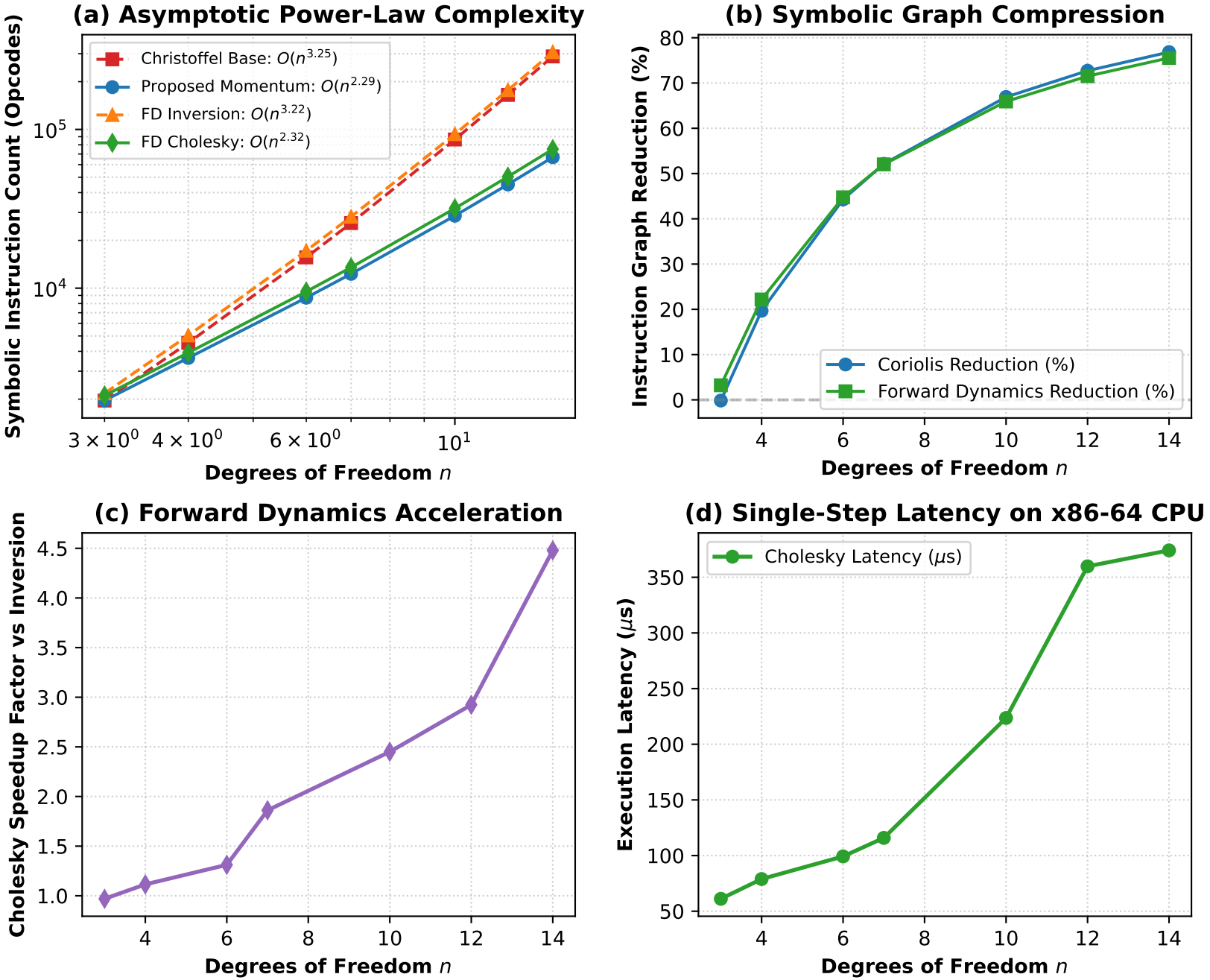
(a) Asymptotic Power-Law Complexity
Symbolic Instruction Count (Opcodes)
Degrees of Freedom n
Christoffel Base: $O(n^{3.25})$
Proposed Momentum: $O(n^{2.29})$
FD Inversion: $O(n^{3.22})$
FD Cholesky: $O(n^{2.32})$
10^5
10^4
3×10^0
4×10^0
6×10^0
10^1
(b) Symbolic Graph Compression
Instruction Graph Reduction (%)
Degrees of Freedom n
Coriolis Reduction (%)
Forward Dynamics Reduction (%)
(c) Forward Dynamics Acceleration
Cholesky Speedup Factor vs Inversion
Degrees of Freedom n
(d) Single-Step Latency on x86-64 CPU
Execution Latency (μs)
Degrees of Freedom n
Cholesky Latency (μs)